

TAI NGUYEN XUAN

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EDUCATION

- **University of Science and Technology of Hanoi (USTH)** Oct 2022 – Aug 2026
Bachelor in Advanced Material Science and Nanotechnology (AMSN)
 - Class Monitor
 - GPA: 17.01/20

RESEARCH EXPERIENCE

Laboratory of Advanced Materials and Application

University of Science and Technology of Hanoi

- **Fabrication and Characterization of Polydopamine-based Memristive Devices** Sep 2025 – Present
Research Assistant | Supervisor: Dr. Nguyen Van Quyen
 - Fabricated Polydopamine (PDA) thin-film crossbar arrays via electrodeposition and electron-beam physical vapor deposition.
 - Characterized film morphology and electrochemical properties using UV-Vis, Raman spectroscopy, and AFM.
 - Developed automated LabVIEW routines to evaluate device reliability and switching kinetics.
- **Visiting Research Scholar** June 2026
Molecular Electronics Lab, Guangdong Technion Israel Institute of Technology, China
 - Evaluated short-term plasticity (STP), paired-pulse facilitation (PPF), and endurance of volatile PDA-based memristors.
 - Conducted XPS analysis of PDA films to examine surface composition and bonding states.
- **Design and Fabrication of a Low-Temperature Cryostat System** Nov 2025 – Mar 2026
Project Group Leader | Supervisor: Dr. Nguyen Van Quyen
 - Designed a custom low-temperature cryostat system utilizing Python-based CAD automation with FreeCAD and AutoCAD.
 - Developed custom multi-layer PCBs via KiCAD for low-temperature electrical characterization.
 - Selected cryogenic-compatible materials for structural components based on thermal conductivity and high-vacuum compatibility.
 - Developed Python-based FDM simulations to model transient heat transfer and optimize cooling.
- **Simulation of Plasmonic Structures** Sep 2024 – Jun 2025
Research Assistant | Supervisors: Dr. Hoang Thi Hong Cam, Assoc. Prof. Nguyen Van Quynh
 - Optimized Au and Au@Ag nanosphere array configurations on SiO_2 via Lumerical FDTD.
 - Modeled transient and steady-state heat transport in plasmonic structures via Lumerical HEAT.

HONORS AND AWARDS

- **Merit Scholarship, USTH** 2023-2025
Awarded to the top 1% of students for outstanding academic excellence.
- **Shaping the Future Initiative Scholarship, USTH** 2025-2026
Awarded to select students for exceptional academic and research achievements.
- **Rector's Award of Excellence, USTH** 2022-2025
Highly selective annual award presented to only 10 students university-wide for remarkable contributions to student leadership and institutional development.
- **Internship Scholarship, USTH** 2025-2026
Competitive funding for high-potential undergraduate research internships.

ADDITIONAL TRAINING

- **Summer School Photovoltaic Workshop** June 2025
Department Electrical Engineer, Tatung University, Taiwan
 - Acquired hands-on technical expertise in PV microgrid architecture, AIoT-driven smart energy systems, and carbon footprint verification methodologies through 10 advanced academic lectures and intensive laboratory sessions.

SKILLS

- **Materials Synthesis & Fabrication:** Photolithography, E-beam Evaporation, E-beam Physical Vapor Deposition (PVD), Electrodeposition, Thin-film Processing
- **Materials Characterization:** X-ray Photoelectron Spectroscopy (XPS), Atomic Force Microscopy (AFM), Surface-Enhanced Raman Spectroscopy (SERS), Raman Spectroscopy, UV-Visible Spectroscopy, Fourier-Transform Infrared Spectroscopy (FTIR), Optical Microscopy
- **Device Testing & Instrumentation:** Fast-pulse Electrical Characterization, IV Sweep Diagnostics, LabVIEW System Automation (Endurance, Retention, Switching Kinetics)
- **Simulation & Computational Modeling:** Ansys Lumerical (FDTD, HEAT), Finite Difference Method (FDM), COMSOL Multiphysics, Abaqus (FEA)
- **Computer-Aided Design (CAD):** CATIA, FreeCAD, AutoCAD, Blender, KiCAD (PCB Design)
- **Programming & Automation:** Python (NumPy, Matplotlib, Pandas, Simulation API Integration), Scientific Computing, Automation Scripting, Git/GitHub, R, LaTeX
- **Languages:** English (IELTS 6.5), French (B1), Vietnamese (Native)

TEACHING & LEADERSHIP EXPERIENCE

- **Academic Teaching & Mentoring** 09/2023 - 03/2026
Tutor & Peer Mentor
 - Provided core academic instruction and structured tutoring in fundamental Physics and Chemistry for undergraduate freshers and peers.
- **USTH Science Art Club** 07/2024 - 02/2025
Club President
 - Directed STEM experiential learning programs, providing physics and chemistry instructional support for visiting high school students.
 - Facilitated scientific literacy initiatives by training club members in research article analysis and academic literature review methodologies.
 - Managed student teams in fabricating interactive STEM exhibition prototypes to promote scientific outreach on campus.
- **USTH Student Association** 03/2023 - 06/2024
Marketing Committee Member
 - Coordinated major campus-wide student initiatives, managing stakeholder communication across the Office of Marketing and Student Affairs.
 - Served as a keynote speaker at time management workshops for incoming freshers.